FROM BC(F, N, Q-1) ONE DEDUCES INJECTIVITY OF THE Q-TH BASE CHANGE MAP AND A DESCRIPTION OF ITS COKERNEL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the choice of the injective embedding, the diagram of the two long exact sequences with the vanishing in the upper right corner, the use of BC(f, n, q-1) for the left vertical arrows, and the refinement of the embedding for part (2).

Editorial note. The label lemma-base-change-q-injective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement is delivered verbatim with its displayed diagram and the three enumerated conclusions; the single cited input is kept as an external pointer with its canonical Stacks tag.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The base change condition BC(f, n, q) and the standing remark. Same chapter, the remark collecting the base change conditions for the fixed morphism f and integer n, with the diagram of schemes used in the statement. Supplies the hypothesis BC(f, n, q-1) and the notation of the diagram.
- Long exact cohomology sequences for the two rows. Same chapter, the long exact sequences of the two rows obtained from a short exact sequence of sheaves annihilated by n. Supplies the exact rows of the displayed diagram, including the vanishing in the upper right corner for an injective I.

Source reference. Original reference 0F02: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 16848–16848. Prerequisite pointer invoked at proof line(s) 16870; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *With $f : X \to S$ and n as in Remark 0F02 assume for some $q \geq 1$ we have $BC(f, n, q-1)$. Then for every commutative diagram*

$$\begin{array}{ccccc} X & \longleftarrow & X' & \xleftarrow[h]{} & Y \\ {\scriptstyle f}\downarrow & & {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle e} \\ S & \longleftarrow & S' & \xleftarrow{g} & T \end{array}$$

with $X' = X \times_S S'$ and $Y = X' \times_{S'} T$ and g quasi-compact and quasi-separated, and every abelian sheaf $\mathcal{F}$ on $T_{étale}$ annihilated by n

(1) *the base change map $(f')^{-1}R^q g_*\mathcal{F} \to R^q h_* e^{-1}\mathcal{F}$ is injective,*
(2) *if $\mathcal{F} \subset \mathcal{G}$ where $\mathcal{G}$ on $T_{étale}$ is annihilated by n, then*

$$\text{Coker}\left((f')^{-1}R^q g_*\mathcal{F} \to R^q h_* e^{-1}\mathcal{F}\right) \subset \text{Coker}\left((f')^{-1}R^q g_*\mathcal{G} \to R^q h_* e^{-1}\mathcal{G}\right)$$

(3) *if in (2) the sheaf $\mathcal{G}$ is an injective sheaf of $\mathbf{Z}/n\mathbf{Z}$-modules, then*

$$\text{Coker}\left((f')^{-1}R^q g_*\mathcal{F} \to R^q h_* e^{-1}\mathcal{F}\right) \subset R^q h_* e^{-1}\mathcal{G}$$

Proof. Choose a short exact sequence $0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{Q} \to 0$ where $\mathcal{I}$ is an injective sheaf of $\mathbf{Z}/n\mathbf{Z}$-modules. Consider the induced diagram

$$\begin{array}{ccccccc} (f')^{-1}R^{q-1}g_*\mathcal{I} & \longrightarrow & (f')^{-1}R^{q-1}g_*\mathcal{Q} & \longrightarrow & (f')^{-1}R^q g_*\mathcal{F} & \longrightarrow & 0 \\ \cong\downarrow & & \cong\downarrow & & \downarrow & & \downarrow \\ R^{q-1}h_* e^{-1}\mathcal{I} & \longrightarrow & R^{q-1}h_* e^{-1}\mathcal{Q} & \longrightarrow & R^q h_* e^{-1}\mathcal{F} & \longrightarrow & R^q h_* e^{-1}\mathcal{I} \end{array}$$

with exact rows. We have the zero in the right upper corner as $\mathcal{I}$ is injective. The left two vertical arrows are isomorphisms by $BC(f, n, q-1)$. We conclude that part (1) holds. The above also shows that

$$\text{Coker}\left((f')^{-1}R^q g_*\mathcal{F} \to R^q h_* e^{-1}\mathcal{F}\right) \subset R^q h_* e^{-1}\mathcal{I}$$

hence part (3) holds. To prove (2) choose $\mathcal{F} \subset \mathcal{G} \subset \mathcal{I}$. □